Architectural Justification: Medical Consent Management System

Architecture Selection

The Medical Consent Management System will use a **Layered Architecture**. The design separates the presentation interface, business services, and data storage responsibilities. The component diagram shows the logical services within the business layer; these are proposed design components, not claims about an implemented system.

Reason 1: Separation of responsibilities

The portal handles user interaction, while the Consent Management Service applies the rules for granting, viewing, and revoking consent. Doctor Verification and Medical Records Access are separate responsibilities. This separation makes the consent rules easier to test, maintain, and change without mixing them with interface code.

Reason 2: Supports the functional requirements

The design supports time-bounded consent for selected records (FR-001), viewing active and past permissions (FR-002), immediate revocation (FR-003), doctor notifications (FR-004), and verification of doctors (FR-005). Before medical records are provided, access must be checked against the doctor's verified identity, the records permitted by the patient, the consent status, and its expiry time.

Security advantage

Centralizing authorization decisions in the business services reduces the risk of relying on the user interface alone to protect medical records. The Audit and Notification Service records consent and access events in an append-only audit trail, supporting the traceability requirement in NFR-001. A revocation must invalidate further access immediately.

Performance benefit

Separating consent processing from data access and audit handling makes slow operations easier to identify and optimize. Consent lookups, database queries, and notification delivery can be tuned independently. The design is intended to support NFR-002: 5,000 concurrent users, with at least 95% of consent requests completing in under two seconds; this target would need to be verified through load testing and is not claimed as already achieved.